NIDHAL DALHOUMI

Full-Stack Software Engineer | Web Development · Testing · AI-Assisted Engineering

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PROFESSIONAL SUMMARY

Junior full-stack software engineer (React, TypeScript, Python) with a mathematics and physics background and a first-principles approach to problem-solving. Built and shipped TheExperience (React, TypeScript, FastAPI, pytest, CI) and an internal healthcare platform. Uses Claude, Cursor and ChatGPT for AI-assisted development, debugging and testing, validating outputs through code review and tests.

TECHNICAL SKILLS

Languages & Frameworks:
Python,Java, TypeScript, JavaScript, React.js, Angular, Django, FastAPI , Spring Boot, REST APIs, WebSockets, OOP, SQL

Testing & Quality Engineering:
pytest, API Testing, Postman, Debugging, Automated Testing, GitHub Actions, Jenkins, SonarQube, Docker, Linux, Bash, Git.

Systems & Databases:
PostgreSQL, MySQL, Redis, JVM Performance Diagnostics.

AI-Assisted Development:
Claude, Cursor, ChatGPT — codebase exploration, debugging, test-case generation, implementation assistance and code review.

EXPERIENCE

Medico International Services (Prosthesis Manufacturer), *Software Engineer (End-of-Studies Project) - Digital transformation initiative: Designed and built the company's first internal software platform* 04/2025 – 04/2026 Tunis,Tunisia

- Built a digital healthcare platform with Angular, Django REST Framework, PostgreSQL, and JWT authentication, supporting scheduling and treatment workflows.
- Implemented real-time messaging using Django Channels, WebSockets, and Redis.
- Integrated YOLO-based image analysis, API validation, testing, and debugging to improve reliability.

IC CANADA, *Software Engineering Intern (Remote)* 06/2024 – 09/2024 Remote

- Developed frontend components in Angular and backend REST APIs in Django for an enterprise BI and ERP platform, integrating relational database functionality via SQL.
- Worked on business data flows, API development, and reporting functionality, maintaining consistency across application and database layers.
- Collaborated in standups and planning sessions; performed root-cause debugging and bug fixes.

PROJECTS

TheExperience — Music Discovery & Visual Experience Platform, 2026
React · TypeScript · FastAPI · Async APIs

- Independently built and shipped a full-stack music discovery product integrating Last.fm, MusicBrainz and Cover Art Archive.
- Implemented asynchronous API integrations, caching, rate limiting and graceful failure handling.
- Developed pytest tests covering data normalization, provider integrations, cover art and album search; configured GitHub Actions CI.
- Used AI-assisted debugging and test-case suggestions to investigate integration issues, explore edge cases and improve test reliability.

Kaddem — Colocation & Carpooling Platform, *Spring Boot · Angular · CI/CD · Docker* 2025 – 2025

- Developed user management, matching and reservation workflows; configured Jenkins/GitHub Actions pipelines with automated testing, SonarQube and Prometheus/Grafana monitoring.

EDUCATION

Software Engineering Diploma, *ESPRIT* 08/2022 – 12/2025

Physics & Chemistry Preparatory Studies, *IPEIT* 09/2019 – 06/2022

LANGUAGES

English: Fluent C2 | **French:** B2 | **Spanish:** Basic | **Arabic:** Native

CLUB LEADERSHIP EXPERIENCE

Chairman — IPEIT Club (2021-2022)

- Organized "Road to 5.0," an AI/Industry 5.0 event .

Chairman — DeepFlow Club, ESPRIT (2023–2024)

- Led AI-focused student org; organized "AI for All" and "AI HackFlow 2.0" (City of Sciences Tunis)